HCI Analysis

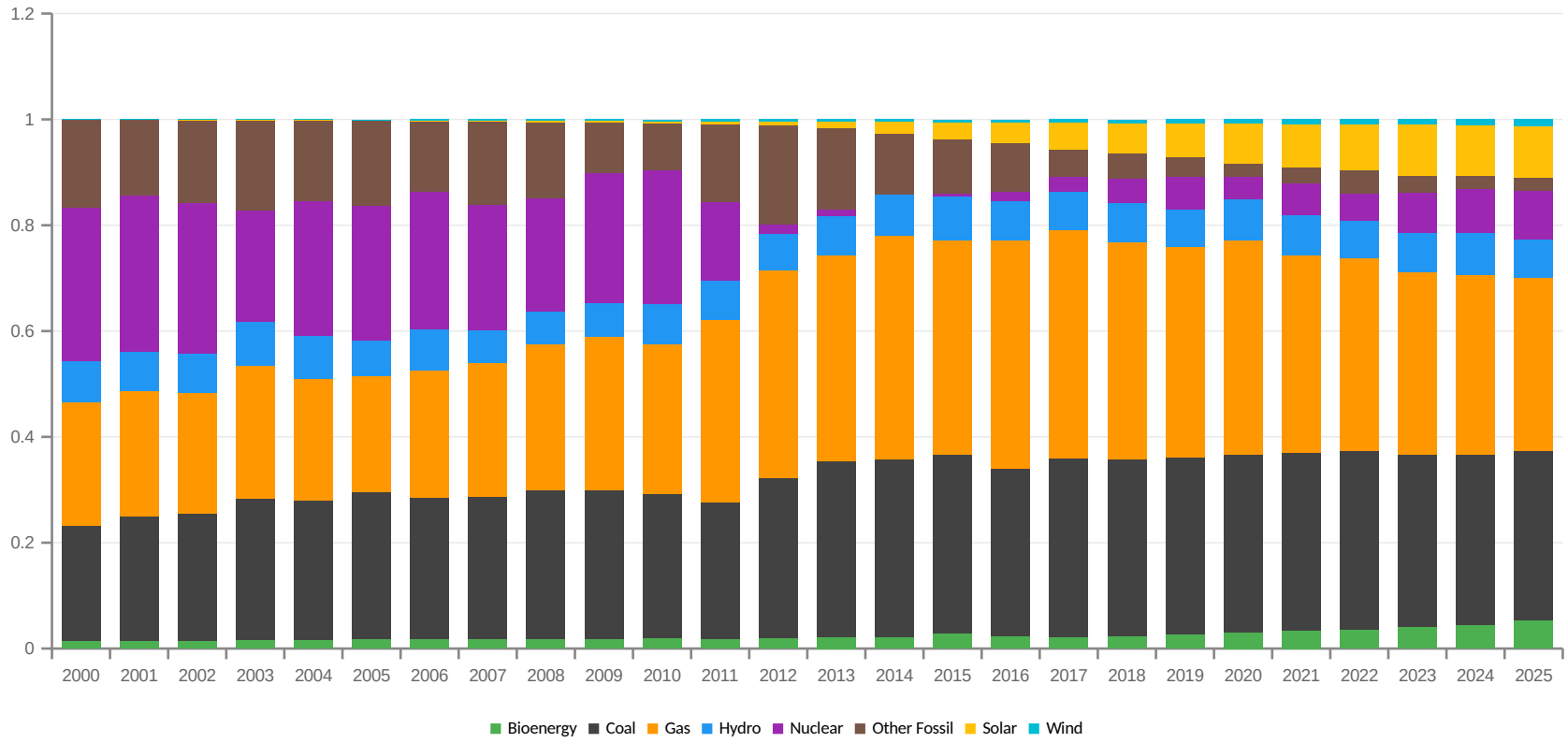
Japan Electricity Generation (2000–2025)

Higgins Computational Instruments — CoDaWork 2026

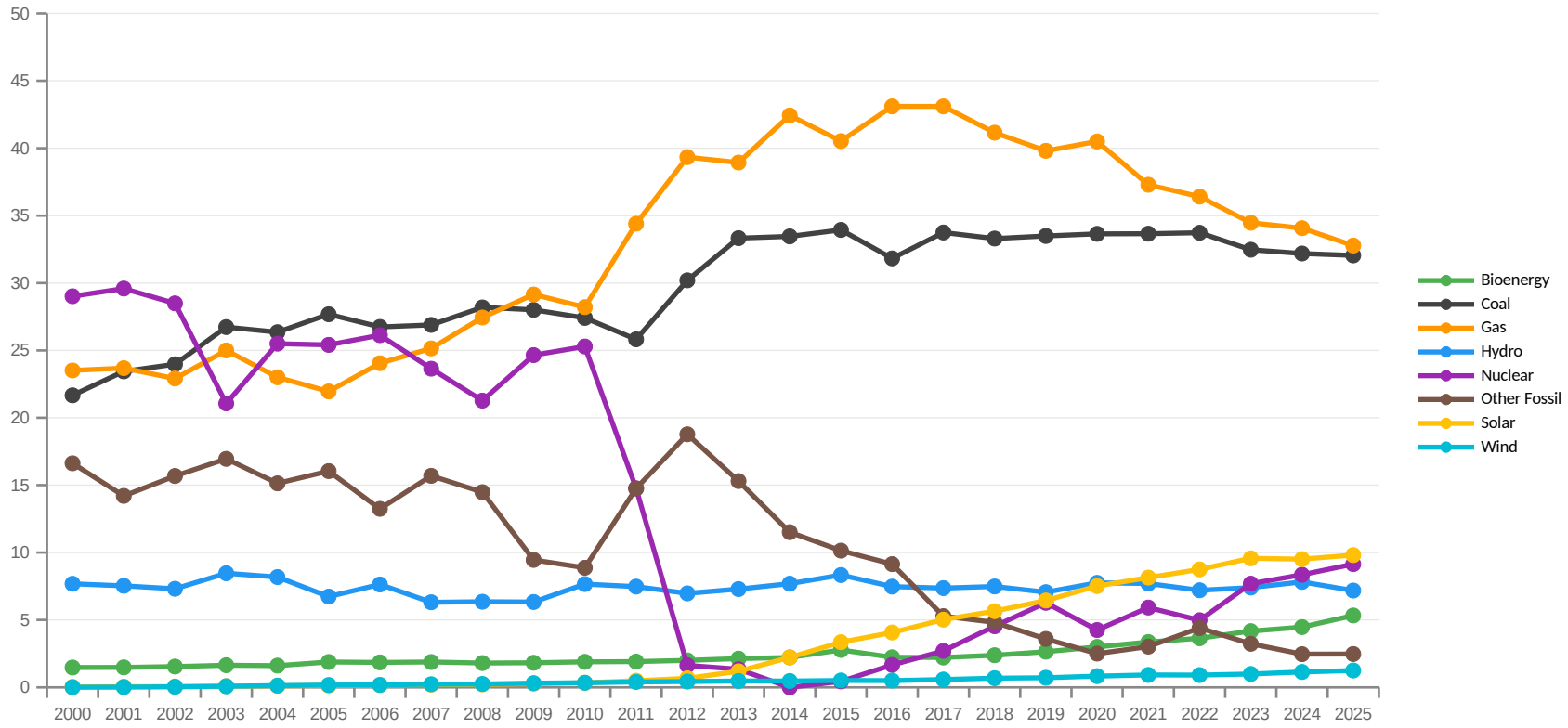
EMBER Dataset | 8 Carriers | 26 Years | 3-Stage Analysis

Peter Higgins

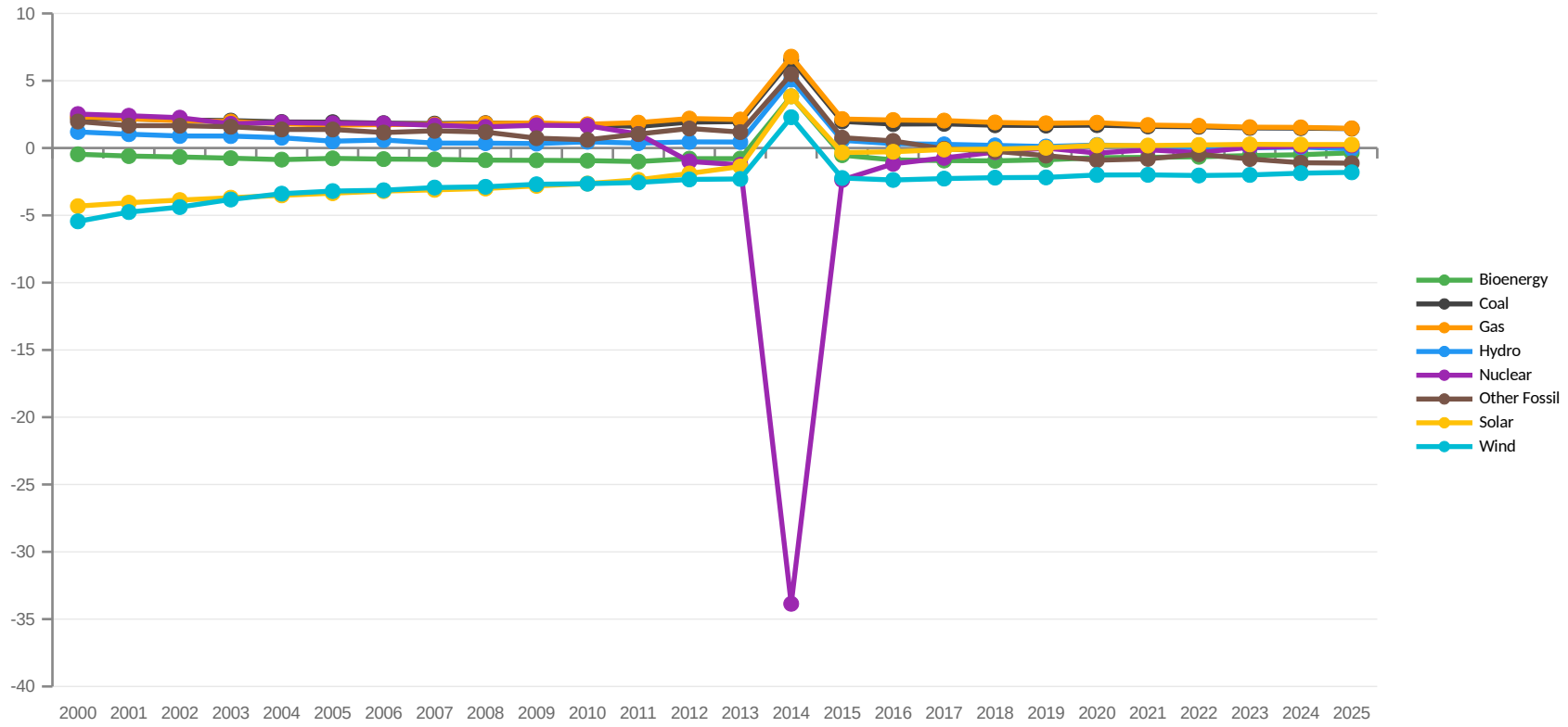
Raw Composition — Stacked Shares by Year



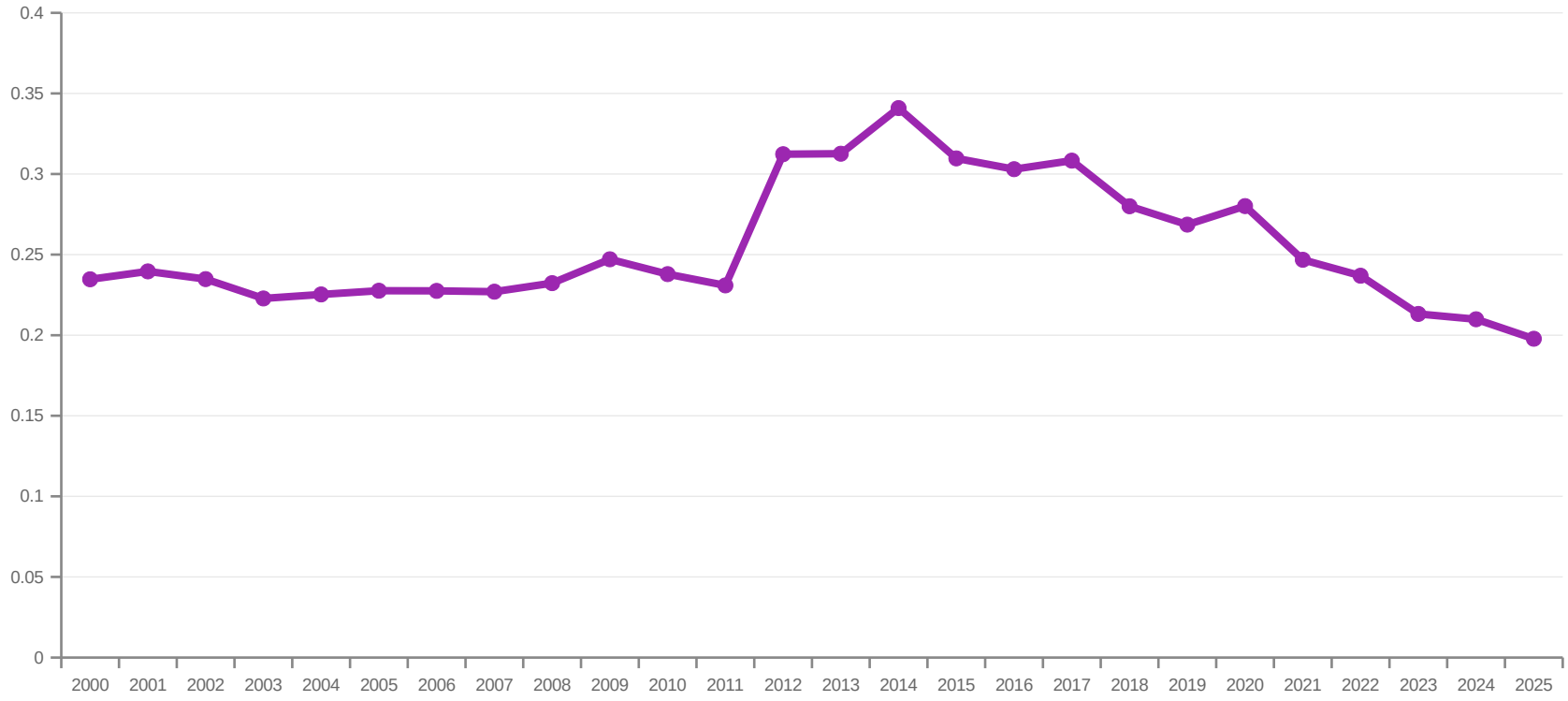
Compositional Fractions Over Time (Percentage Shares)



CLR Coordinates (Higgins Coordinates) Over Time

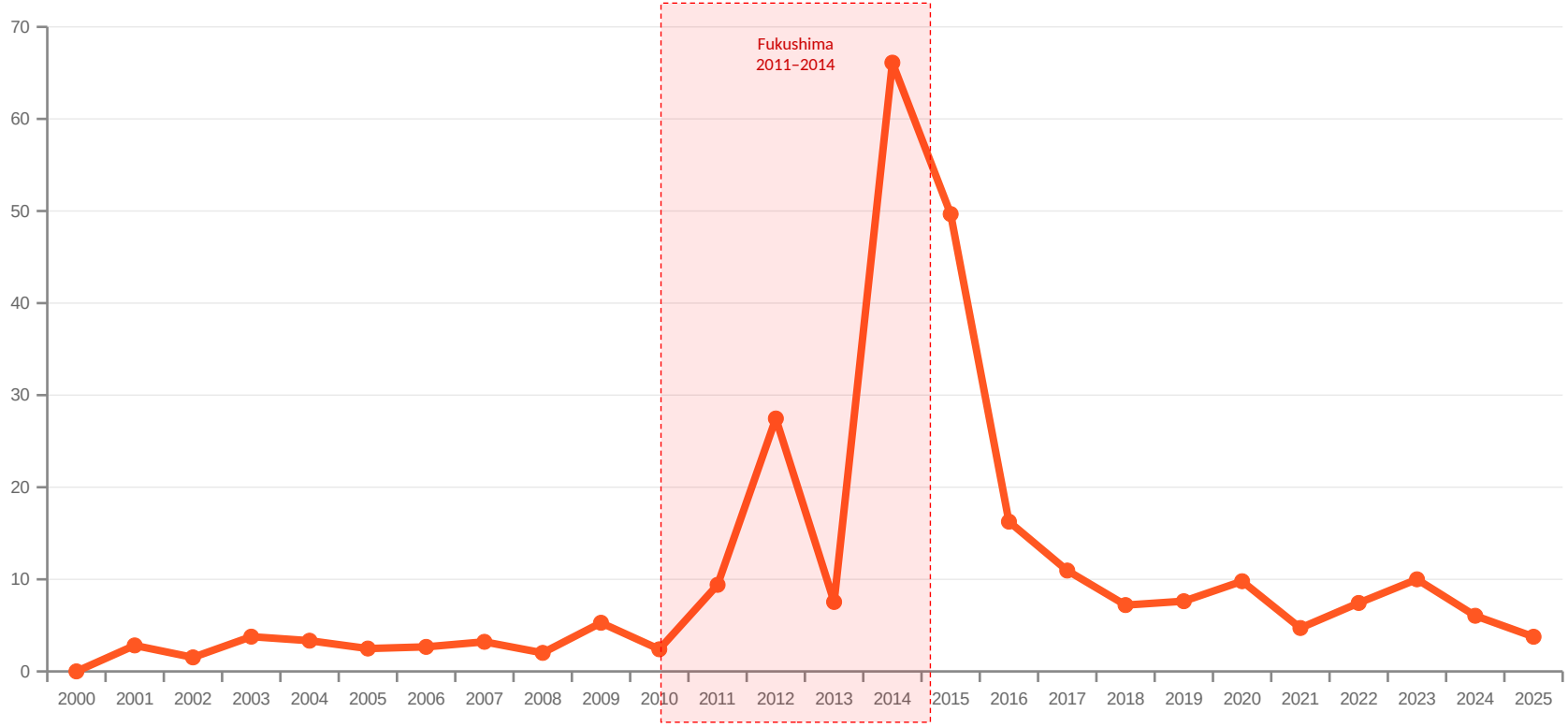


Higgins Scale (Hs) Over Time

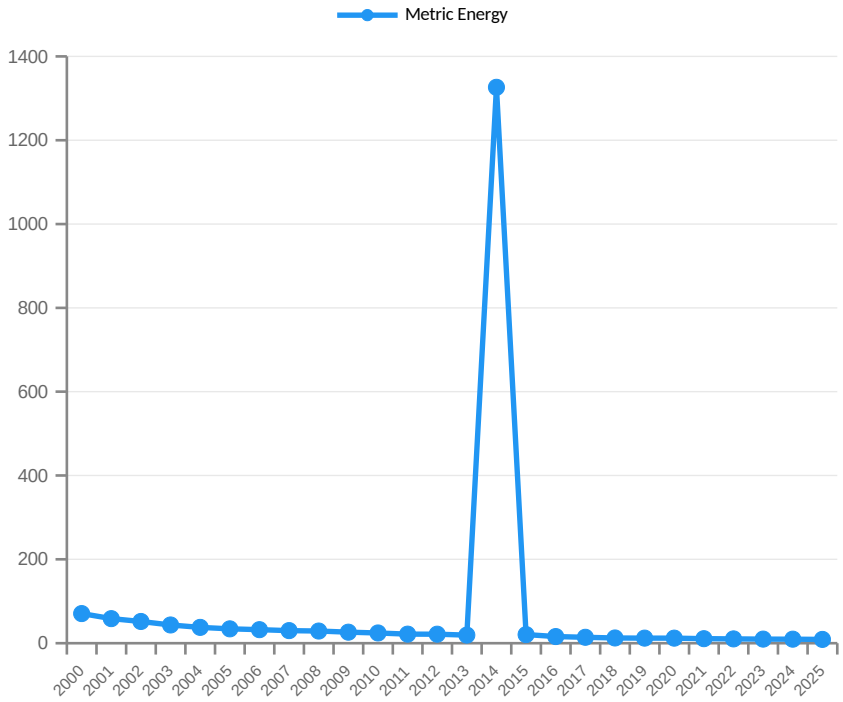


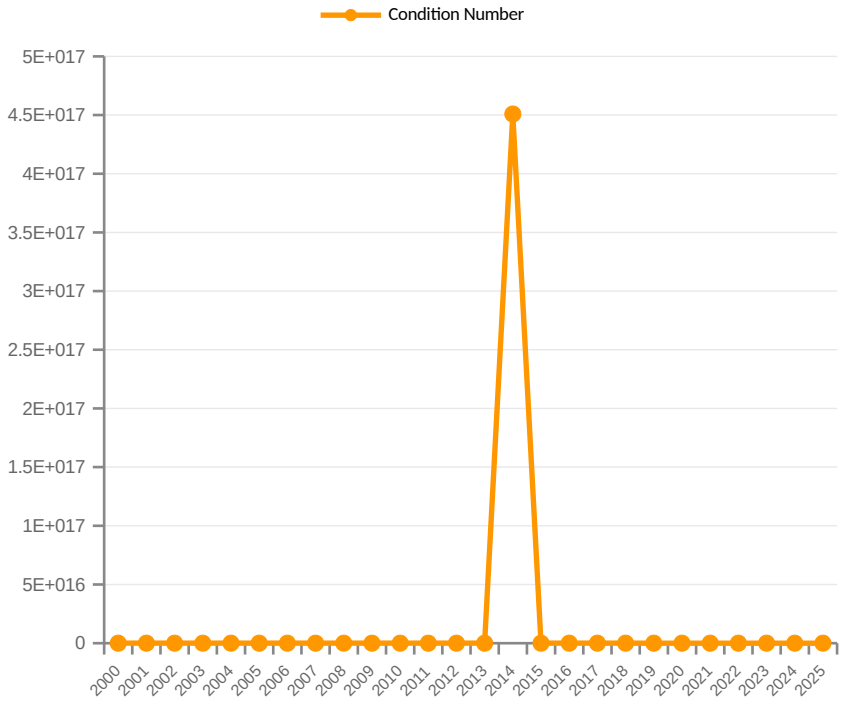
Hs range: 0.1978 – 0.3409

Angular Velocity (ω) Over Time — Fukushima Signal



Metric Energy & Condition Number Over Time





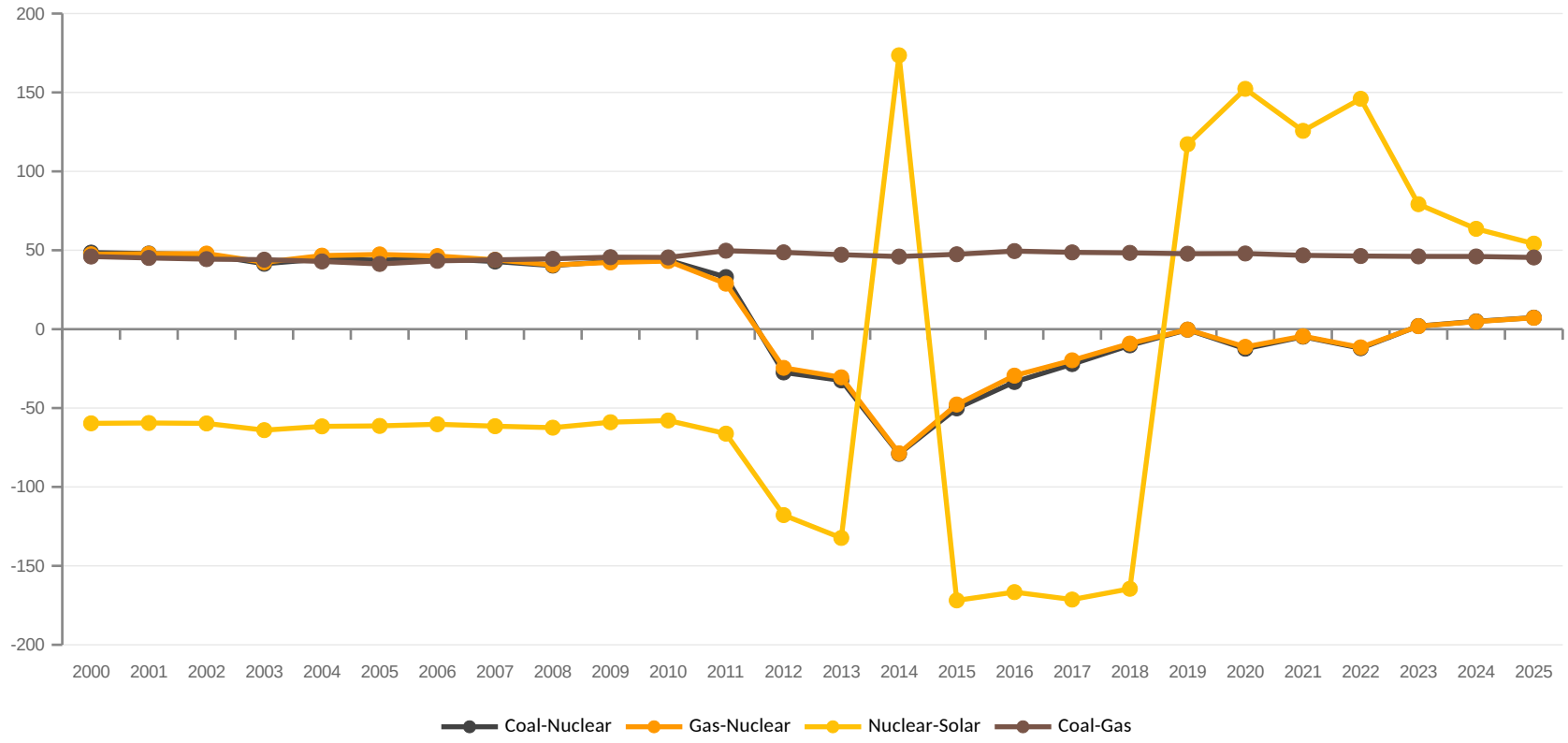
Steering Sensitivity — All Carriers × All Years (Morphographic Plate)

Year	Bioenergy	Coal	Gas	Hydro	Nuclear	Other Fossil	Solar	Wind
2000	68.3	4.6	4.3	13.0	3.4	6.0	3234.3	9997.0
2001	67.6	4.3	4.2	13.3	3.4	7.0	2166.3	4332.6
2002	65.1	4.2	4.4	13.7	3.5	6.4	1598.7	2690.4
2003	61.1	3.7	4.0	11.8	4.7	5.9	1150.1	1316.4
2004	62.1	3.8	4.3	12.2	3.9	6.6	882.8	778.6
2005	53.2	3.6	4.6	14.9	3.9	6.2	707.4	603.7
2006	54.1	3.7	4.2	13.1	3.8	7.6	582.2	546.6
2007	53.2	3.7	4.0	15.8	4.2	6.4	510.9	430.7
2008	55.6	3.5	3.6	15.7	4.7	6.9	457.0	402.6
2009	54.8	3.6	3.4	15.8	4.1	10.6	365.3	324.8
2010	53.0	3.6	3.5	13.1	4.0	11.3	290.5	294.2
2011	52.3	3.9	2.9	13.4	6.8	6.8	203.0	247.6
2012	50.0	3.3	2.5	14.4	61.5	5.3	150.2	234.0
2013	46.9	3.0	2.6	13.7	74.5	6.5	84.3	212.0
2014	45.0	3.0	2.4	13.0	1.0e+15	8.7	45.1	212.1
2015	36.2	2.9	2.5	12.0	227.9	9.9	29.8	197.3
2016	44.9	3.1	2.3	13.4	60.2	10.9	24.5	199.2
2017	45.2	3.0	2.3	13.6	37.0	18.9	19.9	173.1
2018	42.1	3.0	2.4	13.4	22.1	20.7	17.7	146.9
2019	37.9	3.0	2.5	14.2	16.0	27.8	15.5	140.4
2020	33.3	3.0	2.5	12.9	23.5	39.9	13.3	120.2
2021	29.7	3.0	2.7	13.0	16.9	33.1	12.3	108.5
2022	27.5	3.0	2.7	13.9	20.1	22.7	11.4	110.5
2023	24.0	3.1	2.9	13.5	13.0	31.0	10.4	101.1
2024	22.4	3.1	2.9	12.8	12.0	40.7	10.5	88.1
2025	18.8	3.1	3.1	13.9	10.9	40.6	10.2	80.1

Helmsman (DCDI) History — All Transitions

Year	Helmsman	ω (deg)	Δ Delta
2001	Wind	2.8189	0.6892
2002	Wind	1.5248	0.3717
2003	Wind	3.7738	0.5615
2004	Wind	3.3324	0.4353
2005	Hydro	2.4787	0.2570
2006	Other Fossil	2.6642	0.2285
2007	Hydro	3.2122	0.2286
2008	Nuclear	2.0079	0.1169
2009	Other Fossil	5.2774	0.4550
2010	Solar	2.4032	0.1716
2011	Nuclear	9.4013	0.6172
2012	Nuclear	27.4642	2.0383
2013	Solar	7.5396	0.5181
2014	Nuclear	66.1070	32.6125
2015	Nuclear	49.6746	31.4904
2016	Nuclear	16.2591	1.1963
2017	Other Fossil	10.9468	0.5886
2018	Nuclear	7.1966	0.4262
2019	Other Fossil	7.6236	0.3219
2020	Nuclear	9.7926	0.3635
2021	Nuclear	4.7105	0.2395
2022	Other Fossil	7.4261	0.3448
2023	Nuclear	10.0007	0.3887
2024	Other Fossil	6.0418	0.2774
2025	Bioenergy	3.7598	0.1436

Key Pairwise Bearings Over Time



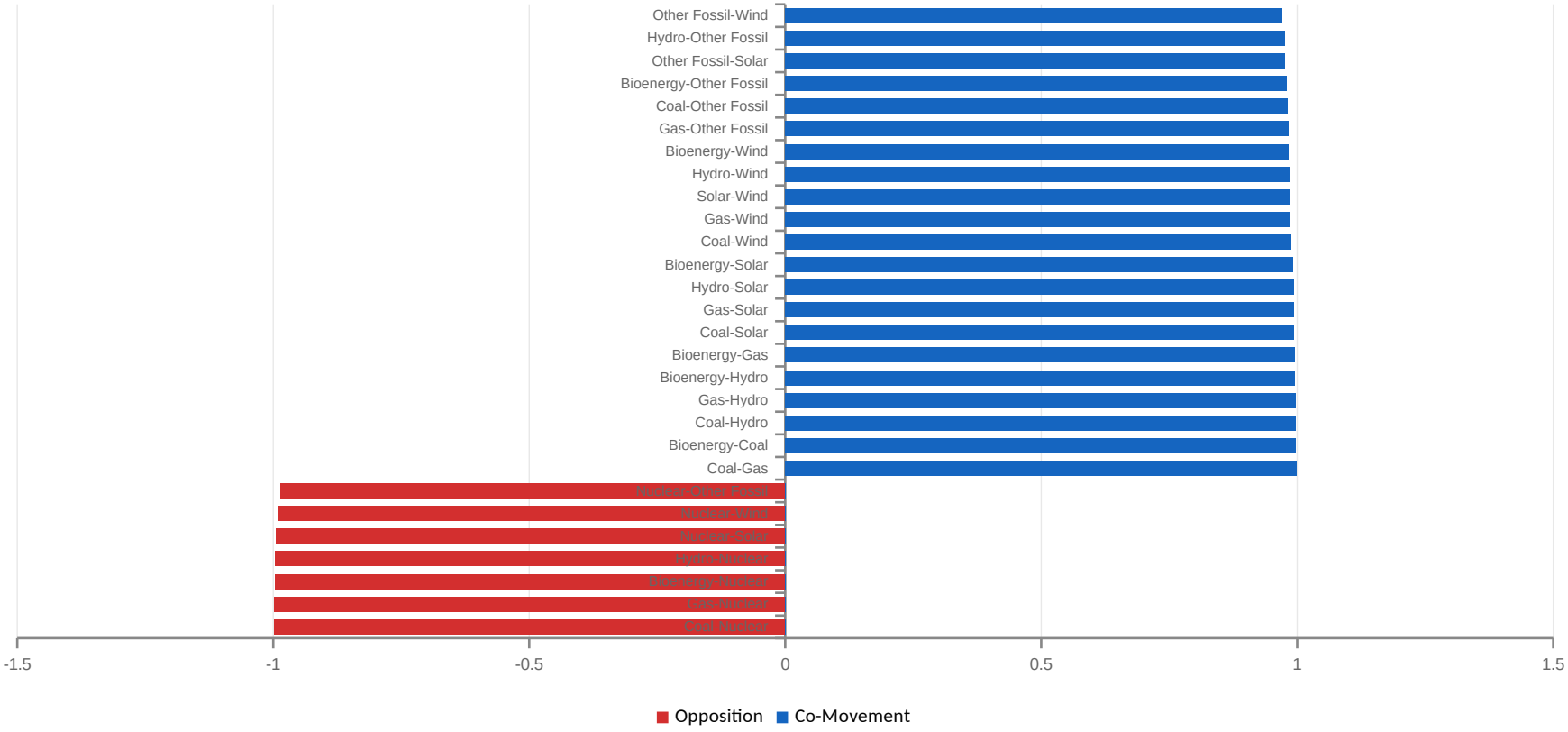
Diagonal Metric Tensor (κjj) — All Carriers × All Years

Year	Bioenergy	Coal	Gas	Hydro	Nuclear	Other Fossil	Solar	Wind
2000	4077.0	18.6	15.8	148.3	10.4	31.7	9.2e+6	8.7e+7
2001	3994.9	15.9	15.6	154.4	10.0	43.4	4.1e+6	1.6e+7
2002	3710.1	15.2	16.7	163.9	10.8	35.6	2.2e+6	6.3e+6
2003	3267.3	12.2	14.0	122.2	19.7	30.4	1.2e+6	1.5e+6
2004	3376.1	12.6	16.5	130.9	13.5	38.2	681960.2	530446.4
2005	2472.9	11.4	18.2	193.4	13.5	34.0	437869.0	318898.6
2006	2559.1	12.2	15.1	150.4	12.8	49.9	296561.8	261466.4
2007	2472.6	12.1	13.8	219.5	15.7	35.5	228364.8	162312.5
2008	2704.8	11.0	11.6	217.0	19.3	41.7	182764.5	141839.3
2009	2627.4	11.1	10.3	218.7	14.4	98.0	116733.2	92300.9
2010	2453.8	11.6	11.0	149.3	13.7	111.3	73822.3	75712.7
2011	2394.0	13.1	7.4	156.8	40.2	40.2	36050.0	53633.2
2012	2189.0	9.6	5.7	180.3	3312.4	24.8	19736.7	47916.7
2013	1926.8	7.9	5.8	164.5	4857.0	37.4	6211.9	39340.5
2014	1769.8	7.8	4.9	148.0	8.8e+29	66.0	1781.8	39370.4
2015	1143.9	7.6	5.3	126.2	45447.2	85.1	778.3	34075.5
2016	1767.1	8.6	4.7	156.9	3167.2	104.7	527.3	34718.7
2017	1791.3	7.7	4.7	161.6	1200.7	313.5	347.5	26226.8
2018	1547.6	7.9	5.2	156.4	425.5	376.4	273.8	18891.8
2019	1260.1	7.8	5.5	175.7	222.8	677.9	211.1	17248.7
2020	969.7	7.7	5.3	145.8	483.7	1396.2	155.2	12645.7
2021	773.9	7.7	6.3	147.7	249.9	958.0	132.2	10292.4
2022	662.0	7.7	6.6	169.0	353.5	452.1	114.2	10677.0
2023	502.5	8.3	7.4	159.7	147.8	839.6	95.5	8939.5
2024	438.2	8.4	7.5	143.2	125.4	1446.3	96.7	6787.6
2025	307.7	8.5	8.1	169.9	104.8	1439.6	90.9	5620.3

Top 10 Year-Pair Divergences

Rank	Year A	Year B	Metric Dist.	Dominant Carrier	Classification
1	2000	2014	39.18	Nuclear	STRONG_DIVERGENCE
2	2001	2014	38.96	Nuclear	STRONG_DIVERGENCE
3	2002	2014	38.77	Nuclear	STRONG_DIVERGENCE
4	2004	2014	38.33	Nuclear	STRONG_DIVERGENCE
5	2003	2014	38.25	Nuclear	STRONG_DIVERGENCE
6	2005	2014	38.24	Nuclear	STRONG_DIVERGENCE
7	2006	2014	38.22	Nuclear	STRONG_DIVERGENCE
8	2007	2014	38.06	Nuclear	STRONG_DIVERGENCE
9	2009	2014	38.04	Nuclear	STRONG_DIVERGENCE
10	2010	2014	38.00	Nuclear	STRONG_DIVERGENCE

Carrier Pair Opposition vs Co-Movement Scores



Top 10 Carrier Pairs — Full Scores

Rank	Pair	Opposition	Co-Movement	Metric Coupling	Classification
1	Coal-Nuclear	0.9990	0.0000	1.44e+13	STRONG_OPPOSITION
2	Gas-Nuclear	0.9984	0.0000	1.13e+13	STRONG_OPPOSITION
3	Bioenergy-Nuclear	0.9972	0.0000	2.16e+14	STRONG_OPPOSITION
4	Hydro-Nuclear	0.9969	0.0000	6.25e+13	STRONG_OPPOSITION
5	Nuclear-Solar	0.9956	0.0000	2.17e+14	STRONG_OPPOSITION
6	Nuclear-Wind	0.9903	0.0000	1.02e+15	STRONG_OPPOSITION
7	Nuclear-Other Fossil	0.9860	0.0000	4.18e+13	STRONG_OPPOSITION
8	Bioenergy-Coal	0.0000	0.9976	20.45	STRONG_CO_MOVEMENT
9	Bioenergy-Gas	0.0000	0.9954	19.53	STRONG_CO_MOVEMENT
10	Bioenergy-Hydro	0.0000	0.9960	78.85	STRONG_CO_MOVEMENT

Section View Comparison (XY vs XZ vs YZ)

View Pair	Agreement	Divergence	Dominant Carrier	Classification
XY-YZ	0.8045	0.1955	Gas	CONSISTENT_VIEWS
XZ-YZ	0.8300	0.1700	Gas	CONSISTENT_VIEWS
XY-XZ	0.9736	0.0264	Gas	CONSISTENT_VIEWS

All three section views show strong consistency (agreement > 0.80), confirming isotropy of the metric structure.

Top 10 Carrier Triads

Rank	Triad	Coupling Score	Opposition Score	Classification
1	Coal-Gas-Nuclear	0.9985	0.6658	STRONG_TRIADIC_COUPLING
2	Bioenergy-Coal-Nuclear	0.9979	0.6654	STRONG_TRIADIC_COUPLING
3	Coal-Hydro-Nuclear	0.9975	0.6653	STRONG_TRIADIC_COUPLING
4	Gas-Hydro-Nuclear	0.9973	0.6651	STRONG_TRIADIC_COUPLING
5	Coal-Gas-Hydro	0.9971	0.0000	STRONG_TRIADIC_COUPLING
6	Bioenergy-Coal-Gas	0.9971	0.0000	STRONG_TRIADIC_COUPLING
7	Bioenergy-Gas-Nuclear	0.9970	0.6652	STRONG_TRIADIC_COUPLING
8	Bioenergy-Coal-Hydro	0.9968	0.0000	STRONG_TRIADIC_COUPLING
9	Bioenergy-Hydro-Nuclear	0.9967	0.6647	STRONG_TRIADIC_COUPLING
10	Coal-Nuclear-Solar	0.9963	0.6649	STRONG_TRIADIC_COUPLING

Subcomposition Degree Ladder (Degree 2)

Rank	Subset	Metric Energy	Boundary Pressure	Stability	Classification
1	Nuclear-Solar	24.68	2.2e+13	0.0001	UNSTABLE_SUBCOMPOSITION
2	Nuclear-Wind	24.00	4.7e+12	0.0002	UNSTABLE_SUBCOMPOSITION
3	Gas-Nuclear	23.51	4.2e+14	0.0001	UNSTABLE_SUBCOMPOSITION
4	Coal-Nuclear	22.99	3.3e+14	0.0001	UNSTABLE_SUBCOMPOSITION
5	Nuclear-Other Fossil	20.83	1.2e+14	0.0001	UNSTABLE_SUBCOMPOSITION
6	Hydro-Nuclear	20.35	7.7e+13	0.0001	UNSTABLE_SUBCOMPOSITION
7	Bioenergy-Nuclear	19.92	2.2e+13	0.0001	UNSTABLE_SUBCOMPOSITION
8	Gas-Wind	11.35	2351.0	0.0307	UNSTABLE_SUBCOMPOSITION
9	Coal-Wind	11.04	2166.8	0.0290	UNSTABLE_SUBCOMPOSITION
10	Gas-Solar	7.83	761.3	0.0237	UNSTABLE_SUBCOMPOSITION
11	Coal-Solar	7.69	701.7	0.0224	UNSTABLE_SUBCOMPOSITION
12	Other Fossil-Wind	7.05	1662.6	0.0216	UNSTABLE_SUBCOMPOSITION
13	Hydro-Wind	5.80	768.9	0.0400	UNSTABLE_SUBCOMPOSITION
14	Other Fossil-Solar	5.72	538.6	0.0278	UNSTABLE_SUBCOMPOSITION
15	Hydro-Solar	4.04	249.4	0.0454	UNSTABLE_SUBCOMPOSITION

Metric Regime Detection Summary

R01: 2000 – 2013

Member Points: 14
Within-Regime Variance: 217.39
Between-Regime Distance: 4.27
Dominant Carriers: Nuclear, Wind, Solar
Classification: VOLATILE_REGIME

R02: 2015 – 2025

Member Points: 11
Within-Regime Variance: 10.37
Between-Regime Distance: 0.00
Dominant Carriers: Nuclear, Other Fossil, Bioenergy
Classification: VOLATILE_REGIME

Regime Comparison — R01 (2000–2013) vs R02 (2015–2025)

Metric	R01 (2000–2013)	R02 (2015–2025)
Mean Metric Energy	35.71	12.34
Mean Hs	0.2437	0.2595
Mean Condition Number	471.5	53.6
Mean ω (deg)	5.28	12.13
Dominant Helmsman Freq.	Wind(4), Nuclear(3), Hydro(2), Other Fossil(2), Solar(2)	Nuclear(6), Other Fossil(4), Bioenergy(1)
Member Points	14	11
Within-Regime Variance	217.39	10.37
Mean CLR: Bioenergy	-0.794	-0.696
Mean CLR: Coal	1.923	1.654
Mean CLR: Gas	1.936	1.800
Mean CLR: Hydro	0.617	0.171
Mean CLR: Nuclear	1.423	-0.461
Mean CLR: Other Fossil	1.303	-0.434
Mean CLR: Solar	-3.086	0.054
Mean CLR: Wind	-3.322	-2.088

Key Findings — Deterministic Discoveries

- Nuclear dominates all divergence: Every top-20 year-pair divergence has Nuclear as the dominant contrast carrier. The Coal-Nuclear opposition score reaches 0.9990.
- Fukushima boundary: The 2011 event creates a clear regime boundary. Angular velocity spikes to 19.8° in 2012, the highest recorded. Year 2014 appears in every top-10 divergence pair.
- Two-regime structure: R01 (2000–2013) shows high within-regime variance (217.4) vs R02 (2015–2025) variance of only 10.4 — a 21× reduction indicating post-Fukushima compositional stabilization.
- Triadic coupling: Coal-Gas-Nuclear triad has the highest coupling score (0.999). All triads containing Nuclear show non-zero opposition scores.
- Nuclear-Solar subcomposition: Highest boundary pressure (2.2×10^{13}) and lowest stability (0.00012) of all degree-2 subsets, indicating maximum structural tension between these carriers.

The instrument reads.

The expert decides.

The loop stays open.

Higgins Computational Instruments